

- 6 (a) As the rod falls, the area enclosed by the rod and the frame decreases and the flux linkage decreases. By Faraday's law, an e.m.f. is induced. Since the circuit is closed, there is an induced current. There will be a magnetic force acting on the current-carrying rod as it is perpendicular to the magnetic field.
-
- (b) The downward motion of the rod causes change in the magnetic flux linkage in the circuit. According to Lenz's law, the magnetic force acting on the induced current must oppose this downward motion.

Hence the magnetic force is upwards.

Note: one can also work out first the direction of the current flow using Lenz's law, then the direction of the magnetic force using FLHR.

(c) magnetic force = $BIL = B \times \frac{\text{e.m.f.}}{R} \times L = B \times \frac{BLv}{R} \times L = \frac{B^2 L^2 v}{R}$

The highest speed is reached when the upward magnetic force and weight of the rod are equal in magnitude.

$$F_B = W$$

$$BIL = mg$$

$$\frac{B^2 L^2 v}{R} = mg$$

$$v = \frac{mgR}{B^2 L^2} = \frac{0.15 \times 9.81 \times 2.0}{0.35^2 \times 2.0^2}$$

$$v = 6.01 \text{ m s}^{-1}$$